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And some models were multitasking.

But the models we've worked with so far couldn't produce output based on our requirements.

This is where **instruction tuning** comes into play and bridges the gap.

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Instruction Tuning

Why am I telling you this? Because the paper I'm going to present will primarily use Instruction Tuning.

MULTITASK PROMPTED TRAINING ENABLES ZERO-SHOT TASK GENERALIZATION

Multitask Prompted Training Enables Zero-Shot Task Generalization (T0)

This study is based on a model developed by a large team and named T0.

Multitask Prompted Training Enables Zero-Shot Task Generalization (T0)

In my presentation, I will explain how models can accomplish tasks they have never seen before with a smart training strategy, without reaching enormous sizes.

Research Objective & Hypothesis

The Core Question

The Main Goal

The Efficiency Challenge

Keywords

The Core Question

Can Zero-Shot Generalization be directly induced by Explicit Multitask Learning?

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Can Zero-Shot Generalization be directly induced by Explicit Multitask Learning?

According to the paper, large models like GPT-3 achieve this indirectly, while T0 aims to do it explicitly.

The Main Goal

To enable a model to perform completely held-out tasks (unseen during training) by training on a massive mixture of prompted datasets.

To improve robustness to prompt wording.

The Efficiency Challenge

Achieving strong performance without massive scale.

Comparison: Outperforming GPT-3 (175B) using a model 16x smaller (11B)

Keywords

Explicit Multitask Learning

Zero-Shot Generalization

Held-out Tasks

Prompt Engineering

Model Efficiency

First, let's explain some concepts. What is Zero-Shot Generalization?

The ability to perform a task that has never been seen or demonstrated during training.

We can think of the basic logic as making inferences like humans do. For example

Imagine you've never seen a **zebra** before.

However, I am giving you this information:

"A zebra is an animal that looks like a horse but has black and white stripes."

First, let's explain some concepts. What is Zero-Shot Generalization?

The ability to perform a task that has never been seen or demonstrated during training.

It produced output with specific inputs.

Some models performed a single task.

And some models were multitasking.

And you already know the concepts of “Striped” and “Horse.”

If I showed you a picture of a zebra later, even if you had never seen one before, you would say, “That’s a zebra.”

Held-Out Dataset

A **held-out dataset** is a portion of data that is deliberately set aside and never shown to the machine learning model during its training.

Held-Out Dataset

However, the held-out data set also allows us to obtain information about the model's zero-shot generalization.

Held-Out Dataset

For example, since the zebra in our previous assumption was not shown before, we can consider it as included in the held-out dataset.

Let’s go back to the paper I’m presenting today

The Status Quo: Implicit Multitask Learning

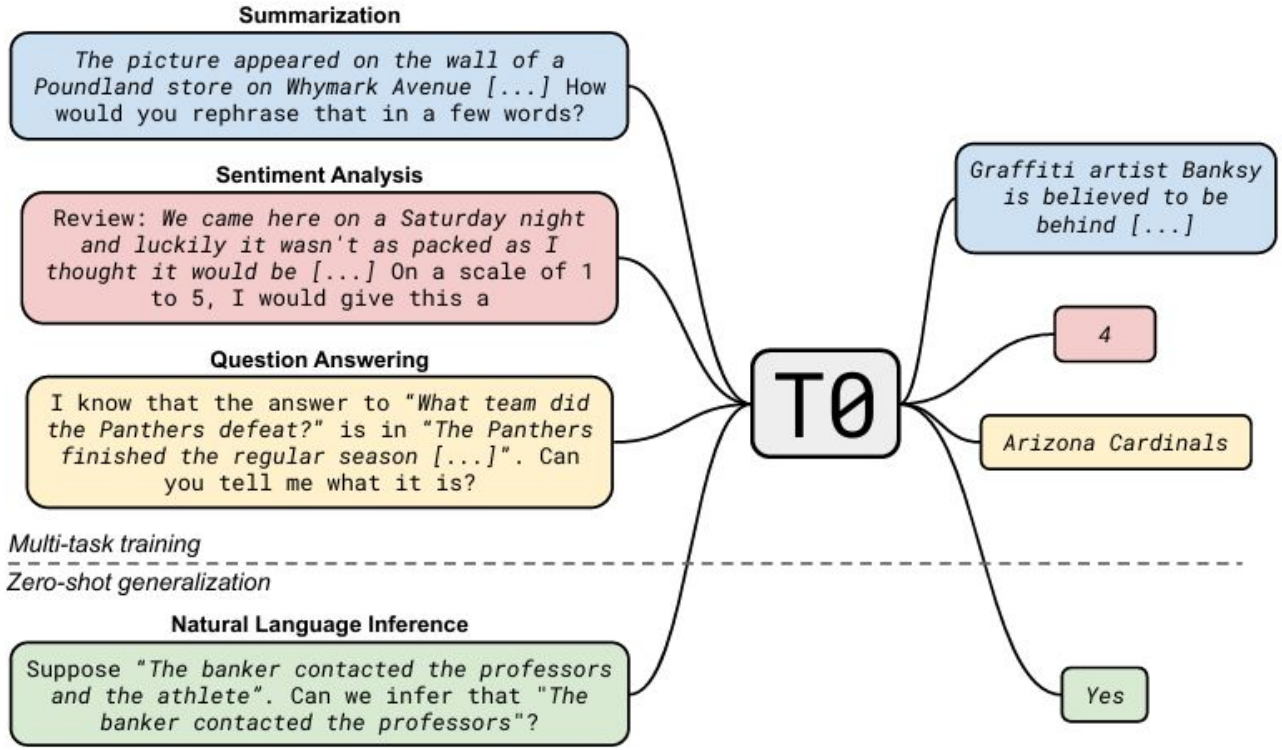
Large language models such as GPT-3 are trained solely on predicting the next word (language modeling).

Hypothesis: When trained on massive data (the internet), these models learn tasks such as question-answering or summarization “by chance” (in an implicit way).

Example: While reading an internet forum, the model observes people asking and answering questions and learns to answer questions “indirectly.”

Problem: This capability requires massive scale and trillions of data points.

Our Approach: Explicit Multitask Learning



Our Approach: Explicit Multitask Learning

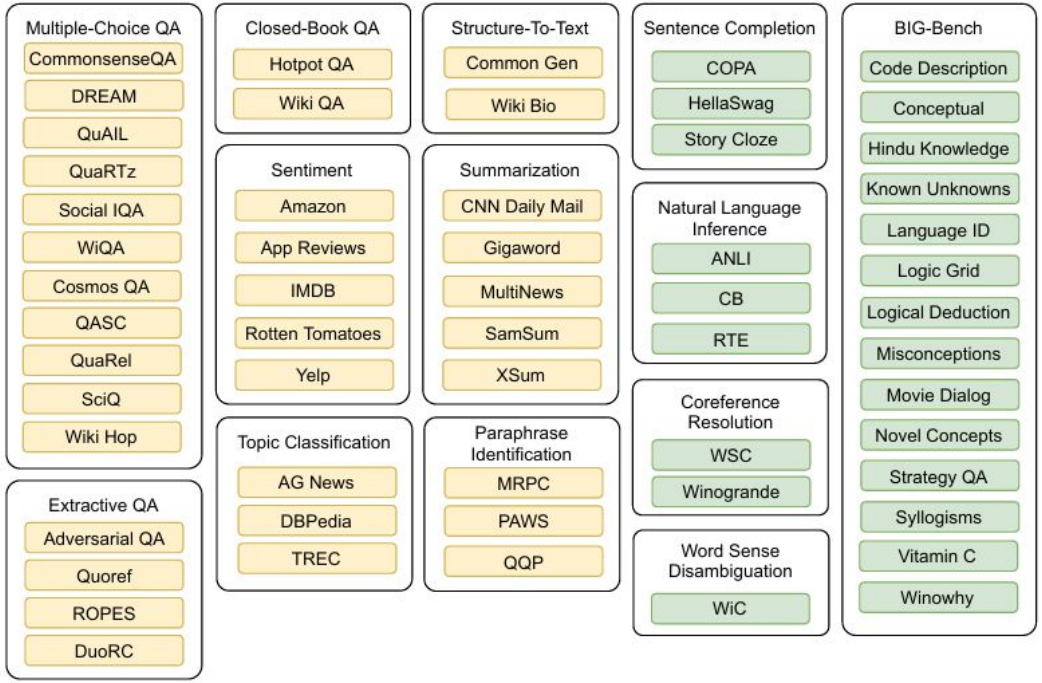
Question: What happens if, instead of leaving the models' tasks to “chance,” we explicitly teach them what to do?

Method: Performing large-scale supervised multi-task training using natural language prompts.

Objective: To enable the model to succeed in tasks it has never encountered before (zero-shot) without scaling it up to massive sizes.

Difference: Instead of waiting for the model to find the data “by chance,” presenting the data to the model in “Prompt” format.

Task Taxonomy: Grouping by Format



T0 datasets and task taxonomy. Color represents the level of supervision. Yellow datasets are in the training mixture. Green datasets are held out and represent tasks that were not seen during training. Hotpot QAis recast as closed-book QA due to long input length.

Task Taxonomy: Grouping by Format

It is difficult to categorize NLP tasks (e.g., is “Common Sense” a skill or a task?).

Decision: Tasks were grouped according to input/output format rather than the skills they require.

Clear Separation: The Training (Yellow) and Test (Green) sets are separated by clear lines..

Fair Comparison: Data that may have leaked into GPT-3's training set was analyzed and removed.

Evaluating Zero-Shot: The 4 Held-Out Task Categories

Natural Language Inference (NLI): The logical relationship between two sentences (Implication, Contradiction). It is a very natural inference ability for humans.

Coreference Resolution: Determining which noun pronouns (it, this, that) refer to.

Sentence Completion: Completing an incomplete sentence logically.

Word Sense Disambiguation: Distinguishing the meaning of a word based on its context in a sentence.

A Unified Prompt Format

From Data to Language: Transforming traditional NLP datasets (X, Y) into natural language instructions.

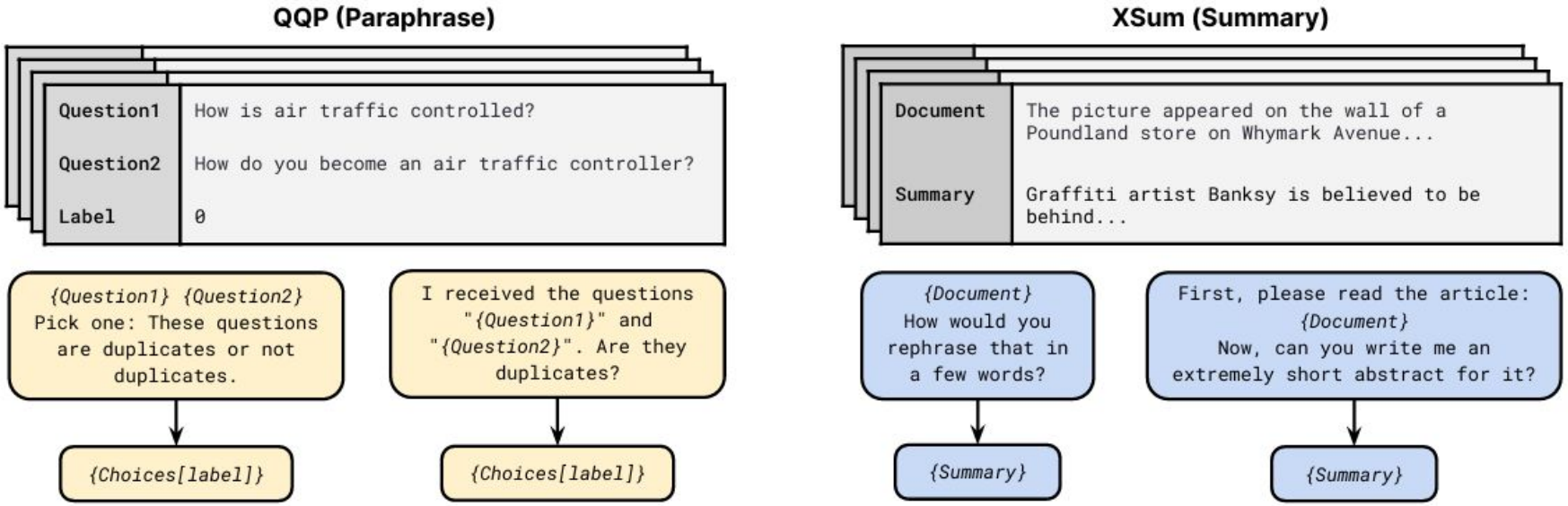
Template Structure: Every prompt consists of two parts:

- Input Template: Maps the data example (e.g., a review) into a question or instruction.
- Target Template: Defines the expected text response.

The Tool: “PromptSource” A templating system developed to facilitate creating massive amounts of prompts.

Diversity: Multiple templates per dataset to teach the model different ways of asking the same thing.

A Unified Prompt Format



Experimental Setup: Model & Training Details

The Model (T0):

Architecture: **Encoder-Decoder** (based on T5).
Base Model: **LM-Adapted T5 (T5+LM)** - Optimized for text generation, not just masking.
Size: **11 Billion Parameters** (Compare to GPT-3's 175B)

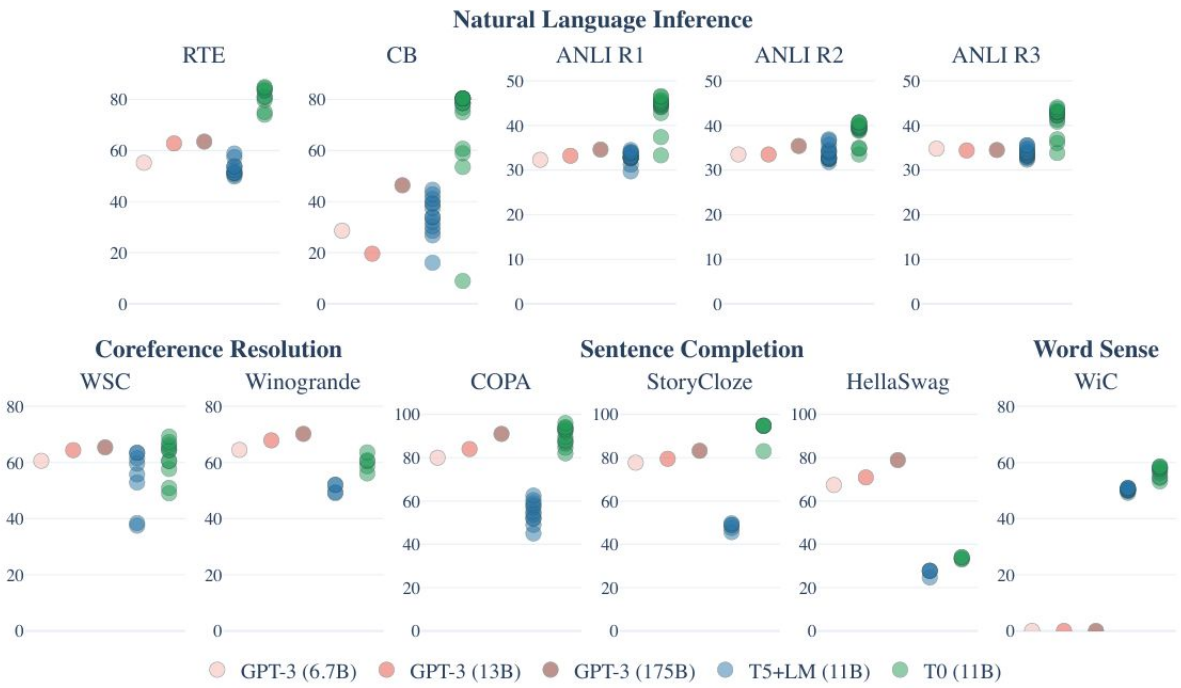
Training Strategy:

Input: A massive multitask mixture of prompted datasets.
Sampling: Proportional sampling to balance large and small datasets.
Checkpointing: Selected based on training task performance (Strict Zero-Shot: No peek at held-out tasks).

Model Variants:

T0: Trained on P3 core tasks.
T0+ / T0++: Trained on additional datasets (like GPT-3 eval sets) to test the limit of generalization.

Results I: Zero-Shot Generalization (T0 vs. GPT-3)



Results I: Zero-Shot Generalization (T0 vs. GPT-3)

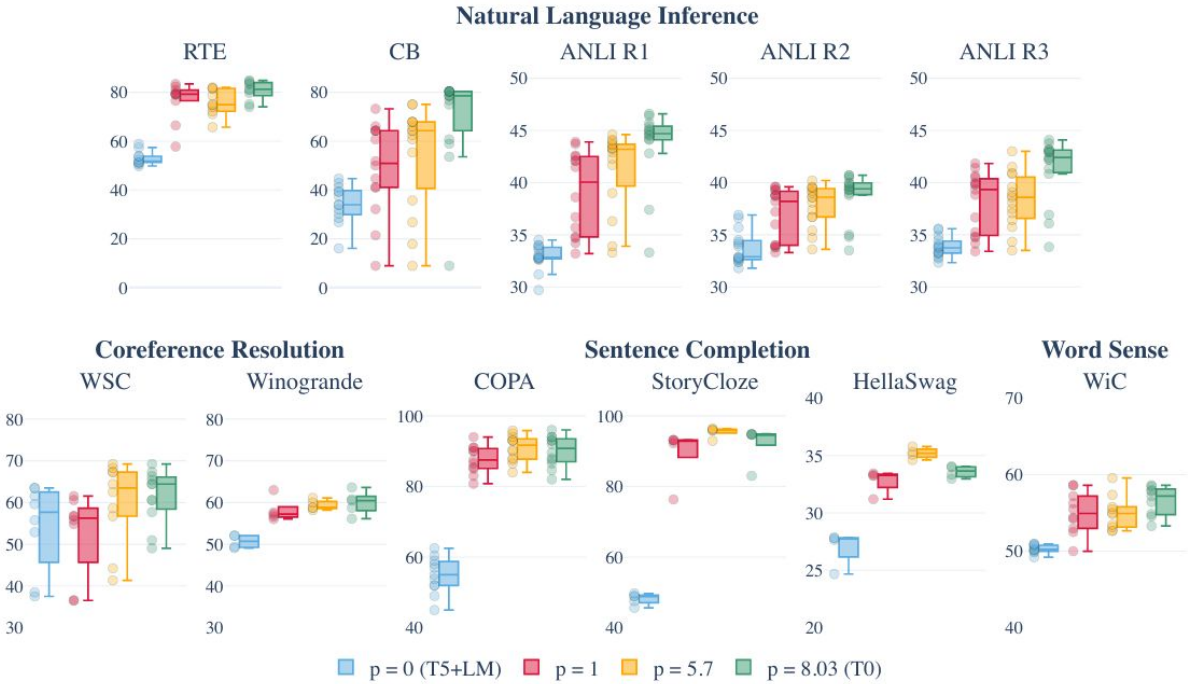
Comparison: Evaluated T0 (11B parameters) against GPT-3 (up to 175B parameters) on 4 held-out tasks.

Key Finding: T0 matches or exceeds the performance of GPT-3 on 9 out of 11 held-out datasets

Efficiency: T0 achieves this despite being **16x smaller** than GPT-3.

NLI Performance: T0 significantly outperforms GPT-3 on Natural Language Inference (NLI) tasks (RTE, CB, ANLI), even though NLI was unseen during training.

Results II: The Power of Prompt Diversity



Results II: The Power of Prompt Diversity

Research Question: Does training on more prompts per dataset improve robustness?

Experiment: Comparing models trained with varying average prompts per dataset (p):

p=1 (Single prompt)
p=5.7 (Multiple prompts)
p=8.03$ (All available prompts)

Impact on Accuracy: Increasing the number of prompts consistently improves the median performance.

Impact on Robustness: More prompts decrease the variance (interquartile range), making the model **less sensitive** to specific wording.

Comparison & Key Insights

Comparison with FLAN: T0 achieves results competitive with Google's FLAN model despite being **10x smaller** (11B vs. 137B parameters).

Architectural Advantage: Unlike FLAN (Decoder-only), T0 uses an **Encoder-Decoder** architecture (T5) with Masked Language Modeling, which is more data-efficient.

Importance of Diversity: T0's prompts are **qualitatively more diverse** and creative (thanks to PromptSource), enabling better generalization at smaller scales.

The Role of Instructions: For some sentence completion tasks (e.g., HellaSwag), removing explicit instructions improved performance, suggesting that "conciseness" sometimes matters.

Conclusion

Core Achievement: Multitask prompted training is a viable and efficient alternative to unsupervised language model pretraining for zero-shot generalization.

David vs. Goliath: The T0 model demonstrates that smaller models can outperform massive ones (like GPT-3) when trained with diverse prompts.

Key Insight: Prompt diversity and the number of training tasks are critical factors for model robustness and performance.

Community Contribution: Full release of trained models (T0), the prompt collection (P3), and the annotation tool (PromptSource) to foster future research.